CONSTRAINTS

"Constraints are the rules given to a column for validating the data."

Types of Constraints

1. **UNIQUE**
2. **NOT NULL**
3. **CHECK**
4. **PRIMARY KEY**
5. **FOREIGN KEY**

1. **UNIQUE:** It is a constraint assign to a column which should not accept duplicated/repeated values.

   Ex:

	Emp_id
	101
101 ✖	102

2. **NOT NULL:** It is a constraint assign to column which should not accept null values.

	Emp_id
	101
	102
✖	
104	

3. **CHECK:** It is an extra validation with a condition. If the condition is satisfied, then the values is accepted else it rejected.

PH_NO
1234567890
1023456789

12345667 ✖

CHECK(LENGTH(PHNO)=10) ← **Condition**

4. **PRIMARY KEY:** It is a constraint used to uniquely identify records from the table.

Characteristics:

- **We can have only one primary key in a table.**
- **It will not accept duplicated/repeated values**
- **It will not accept Null values.**
- **It is a combination of UNIQUE and NOT NULL constraint.**
- **It is not mandatory but highly recommended.**

5. **FOREIGN KEY:** It is used to establish connection between two tables.

Characteristics:

- **We can have multiple foreign key in a table.**
- **It will accept duplicated/repeated values**
- **It will accept Null values.**
- **It is not a combination of UNIQUE and NOT NULL constraint.**
- **For an attribute to become a foreign key first it should be primary key in its ow table.**
- **It is also known as referential integrity constraints.**

Example for Foreign Key:

Emp

EID	NAME	SALARY	DNO FK	CID FK
1	A	10000	20	2
2	B	20000	10	3
3	C	35000	20	1
4	D	50000	10	2

Child Table

Customer

CID	CNAME	CNO
1	X	1001
2	Y	2002
3	Z	3003

Parent Table

Dept

DNO	DNAME	LOC
10	D1	L1
20	D2	L2

Parent Table